

Probability Distribution: Discrete and Continuous

- Discrete random variables may result from a variety of experimental situations, in engineering and applied science they are often associated with counting.
- If X is a discrete random variable, then $E_X(x)$ will have at most a countably infinite number of values. : Range set $= \{x_1, x_2, \dots, x_k, \dots\}$

Example: A Geiger counter is connected to a gas tube in such a way that it will record the background radiation count for a selected time interval $[0, t]$. The random variable of interest is the count. If X denotes the random variable, then $R_X = \{0, 1, \dots, k, \dots\}$

Definition: If X is a discrete random variable, we associate a number $p(x_i) = P(X=x_i)$ with each outcome x_i in R_X for $i=1, 2, \dots, n, \dots$ such that

(i) $p(x_i) \geq 0$ for all i

(ii) $\sum_{i=1}^{\infty} p(x_i) = 1$

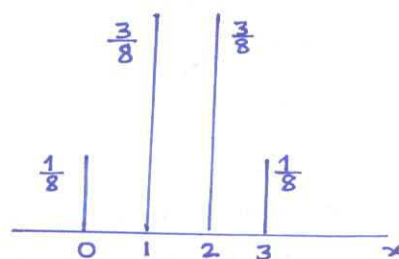
The function $p(x)$ is called the probability mass function (PMF) of the random variable X , and the collection of pairs $[(x_i, p(x_i))]$, $i=1, 2, \dots$ is called the probability distribution of X .

Ex: For the coin-tossing experiment (tossing an unbiased coin thrice), suppose X = the number of heads. The probability distribution is given below:

Tabular Presentation

x	$P(x)$
$P(X=0)$	$1/8$
$P(X=1)$	$3/8$
$P(X=2)$	$3/8$
$P(X=3)$	$1/8$

Graphical Presentation



• The Binomial Distribution: $\text{Bin}(n, p)$

The random variable X that denote the number of successes in n Bernoulli trials has a binomial distribution given by $p(x)$, where

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x=0, 1, 2, \dots, n.$$

$$= 0 \quad \text{otherwise}$$

Ex: A production process produces thousands of parts per day. On the average 1% of the parts are defective and this average does not vary with time. Every hour, a random sample of 100 parts is selected from a conveyor and several characteristics are observed and measured on each part; however, the inspector classifies the part as either good or defective. If we consider the sampling as $n=100$ Bernoulli trials with $p=0.01$, the total number of defectives in the sample x , would have a binomial distribution

$$p(x) = \binom{100}{x} (0.01)^x (0.99)^{100-x}, \quad x=0, 1, 2, \dots, 100.$$

$$= 0 \quad \text{otherwise}$$

Suppose the inspector has instructions to stop the process if the sample has more than two defectives. Find $P(X > 2)$.

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2)\}$$

$$= 1 - (0.99)^{100} + 100(0.01)^1(0.99)^{99} + 4950(0.01)^2(0.99)^{98}$$

$$= 0.08$$

Thus, the probability of the inspector stopping the process is approximately 0.08. The mean number of defectives that would be found is $E(X) = np = 100(0.01) = 1$, and the variance is $V(X) = npq = 0.99$.

• The Poisson Distribution: $P(\lambda)$

A random variable X , taking on one of the values $0, 1, 2, \dots$, is said to be a Poisson RV with parameter λ ($\lambda > 0$), if its probability mass function is given by

$$P(X=i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad i=0, 1, 2, \dots$$

Ex: Suppose that the average number of accidents occurring weekly on a particular stretch of a highway equals 3. Calculate the probability that there is at least one accident this week.

Solⁿ: Let X denote the number of accidents occurring on the stretch of highway during this week. The required probability is

$$P(X \geq 1) = 1 - P(X=0)$$

$$= 1 - e^{-3} \cdot \frac{3^0}{0!}$$

$$= 1 - e^{-3}$$

$$\approx 0.9502.$$

Ex: Consider an experiment that consists of counting the number of α -particles given off in a 1-second interval by 1-gram of radioactive material. If we know from past experience that, on the average, 3.2 such α -particles are given off, what is a good approximation to the probability that no more than 2 α -particles will appear?

S: The number of α -particles given off will be a Poisson RV with parameter $\lambda = 3.2$.

Hence the desired probability is

$$P(X \leq 2) = e^{-3.2} + \frac{3.2}{1!} e^{-3.2} + \frac{(3.2)^2}{2!} e^{-3.2}$$

$$= 0.382$$

• Development of the Poisson Distribution from the Binomial Distribution:

Let $X \sim \text{Bin}(n, p)$. Suppose $n \rightarrow \infty$, $p \rightarrow 0$ such that $np = \lambda$ (a constant), then $p(x) \rightarrow \frac{e^{-\lambda} \lambda^x}{x!}$.

Ex: The probability that a particular rivet in the wing surface of a new aircraft is defective is ~~0.001~~ ^{0.001}. There are 4000 rivets in the wing. What is the probability that not more than six defective rivets will be installed?

S:

$$P(X \leq 6) = \sum_{x=0}^6 \binom{4000}{x} (0.001)^x (0.999)^{4000-x}$$

Using the Poisson approximation, $\lambda = 4000(0.001) = 4$

$$\text{and } P(X \leq 6) = \sum_{x=0}^6 \frac{e^{-4} 4^x}{x!} = 0.889.$$

The Moment Generating Function (MGF)

Def: Given a random variable X , the moment-generating function $M_X(t)$ of its probability distribution is the expected value of e^{tX} . Expressed mathematically,

$$M_X(t) = E(e^{tX})$$

$$= \begin{cases} \sum_i e^{tx_i} p(x_i) & \text{discrete } X, \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & \text{continuous } X \end{cases}$$

Expanding e^{tX} as a power series in t , we obtain

$$e^{tX} = 1 + tX + \frac{t^2 X^2}{2!} + \dots + \frac{t^n X^n}{n!} + \dots$$

On taking expectations,

$$M_X(t) = E(e^{tX}) = 1 + E(X) \cdot t + E(X^2) \cdot \frac{t^2}{2} + \dots + E(X^n) \cdot \frac{t^n}{n!} + \dots$$

$$= 1 + \mu'_1 \cdot t + \mu'_2 \cdot \frac{t^2}{2} + \dots + \mu'_n \cdot \frac{t^n}{n!} + \dots$$

Therefore $\frac{d^n}{dt^n} M_X(t) \Big|_{t=0} = \mu'_n$.

Ex. Suppose that X has a binomial distribution $B(n, p)$. The moment-generating function $M_X(t)$ is

$$M_X(t) = \sum_{x=0}^n e^{tx} p(x)$$

$$= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x}$$

$$\text{or, } M_X(t) = [pe^t + (1-p)]^n.$$

Taking the derivatives, we obtain

$$M'_X(t) = npe^t [1 + p(e^t - 1)]^{n-1} \quad \text{and} \quad M''_X(t) = npe^t (1-p + npe^t) [1 + p(e^t - 1)]^{n-2}.$$

$$\text{Thus } \mu'_1 = \mu = M'_X(t) \Big|_{t=0} = np \quad \text{and} \quad \mu'_2 = M''_X(t) \Big|_{t=0} = np(1-p + np).$$

$$\therefore \text{The variance } \sigma^2 = \mu'_2 - \mu'^2 = np(1-p).$$

Ex: Derive the moment generating function of a Poisson distribution $P(\lambda)$ and hence find the mean and variance.

Ex: The concentration of reactant in a chemical process is a random variable having probability distribution

$$f(r) = 6r(1-r), \quad 0 \leq r \leq 1,$$

$$= 0, \quad \text{otherwise.}$$

The profit associated with the final product is $P = \$1.00 + \$3.00R$. Find the expected value of P . What is the probability distribution of P ?